A Fighting Game AI Using Highlight Cues for Generation of Entertaining Gameplay

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***Abstract*—In this paper, we propose a fighting game AI that selects its actions from the perspective of highlight generation using Monte-Carlo tree search (MCTS) with three highlight cues in the evaluation function. The proposed AI is targeted for being used to generate gameplay in live streaming platforms such as Twitch and YouTube where a large number of spectators watch gameplay to entertain themselves. Our results in a user study conducted using FightingICE, a fighting game platform used in an international game AI competition since 2013, show that gameplay generated by the proposed AI is more entertaining than that by a typical MCTS AI. Detailed analyses of gameplay from all the methods assessed in the user study are also given in the paper.**

***Index Terms*—Monte-Carlo tree search, live streaming, highlight generation, fighting game AI, FightingICE**

I. Introduction

In recent years, live streaming platforms such as Twitch and YouTube are popular. A large group of spectators is "Let's Play" [1] who watch gameplay videos to entertain themselves. As a results, this type of spectators has gained a lot of interests by researchers in various areas.

Recently, Thawonmas and Harada in our group proposed a concept called procedural play generation (PPG) [2]. Their goal is to automatically generate gameplay according to spectators' preferences. PPG requires a system that analyzes and recommends gameplay and a mechanism or AI that generates various kinds of gameplay which entertain different types of spectators. In our previous work on PPG, a method was proposed using Monte-Carlo tree search (MCTS) [3-4] to generate playing styles [5] in a fighting game. We also focus on the AI part for the fighting game genre in this work.

In this paper, inspired by existing highlight generation methods that select exciting scenes for sports spectators, we propose a fighting game AI for generating entertaining gameplay where a combination of highlight indicators or cues is used in the evaluation function of MCTS. We show that gameplay generated by the proposed method is more entertaining than gameplay by a standard MCTS in a user study conducted on a fighting game platform called FightingICE [6], used in international game AI competitions since 2013 including one at CoG 2019. The proposed idea of introducing highlight cues in the evaluation function of MCTS for generation of entertaining gameplay can be applied to other games provided that game forward models are available and cues are modified accordingly.

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II. Related Work

A. Monte-Carlo Tree Search in Fighting Games

Although AIs using deep learning techniques [7-9] can be recently seen, MCTS, combining a Monte-Carlo method and game tree search using a given forward model, is a popular technique to implement a fighting game AI. A sample AI using MCTS for FightingICE [10] adopted the open loop approach [11]. In this work, we follow this recipe.

Figure 1 shows an overview of MCTS. In the open-loop approach, a node represents an action, except for the root node representing the current game state defined by information such as the Hit-Point (HP), energy, coordinates, and action of each character and the game remaining time. An edge represents the ongoing execution of an action of interest. Four steps exist: selection, expansion, simulation, and backpropagation. They are described in the following, respectively.

1) Selection: Nodes are selected from the root node until a leaf node is reached according to a selection criterion in use. We use Upper Confidence Bounds ($UCB1$) [12], which is widely used for this task, defined by the following equation:

$$UCB1_i = \overline{X}_i + C\sqrt{\frac{2\ln N}{N_i}} \quad (1)$$

where N_i is the number of times node (action) i has been visited, N is the number of visits to its parent node, and C is a constant. In addition, $\overline{X}_i$ is the average evaluation value of node i:

$$\overline{X}_i = \frac{1}{N_i}\sum_{j=1}^{N_i} Eval_j \quad (2)$$

where $Eval_j$ is an evaluation function returning a reward value gained in the jth simulation from the perspective of the AI player. Note that every node of the tree contains the $UCB1$ value and a counter counting how many times it has been visited. In this work, the selected path is the one that contains

利用精彩片段线索生成娱乐性游戏过程的格斗游戏人工智能

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摘要——本文提出一种格斗游戏人工智能，该人工智能采用蒙特卡洛树搜索，在评估函数中引入三种精彩片段线索，从精彩片段生成的角度选择动作。所提出的人工智能旨在为 Twitch 和 YouTube 等直播平台生成游戏过程，这些平台上有大量观众通过观看游戏过程获得娱乐。我们使用 FightingICE 开展了用户研究，结果表明，所提出的人工智能生成的游戏过程比典型的蒙特卡洛树搜索人工智能生成的游戏过程更具娱乐性。FightingICE 是一个自2013年起用于国际游戏人工智能竞赛的格斗游戏平台。本文还详细分析了用户研究中所评估的所有方法生成的游戏过程。

索引词—蒙特卡洛树搜索、直播、精彩片段生成、格斗游戏人工智能、FightingICE

I. 引言

近年来，Twitch 和 YouTube 等直播平台广受欢迎。其中一大类观众是“游戏实况” [1]观众，他们通过观看游戏过程视频来娱乐消遣。因此，这类观众引起了各领域研究人员的广泛关注。

最近，我们团队的 Thawonmas 和 Harada 提出了一个名为程序化游戏过程生成（程序化游戏过程生成） [2]的概念。他们的目标是根据观众的偏好自动生成游戏过程。程序化游戏过程生成需要一个分析和推荐游戏过程的系统，以及一种能够生成各类游戏过程、为不同类型的观众带来娱乐的机制或人工智能。在我们此前关于程序化游戏过程生成的研究中，我们提出了一种利用蒙特卡洛树搜索（蒙特卡洛树搜索） [3-4] 在格斗游戏中生成游戏风格 [5] 的方法。本研究同样聚焦于格斗游戏领域的人工智能部分。

本文受现有精彩片段生成方法的启发，提出了一种用于生成娱乐性游戏过程的格斗游戏人工智能，在蒙特卡洛树搜索的评估函数中结合使用多种精彩片段指标或线索；这些现有方法旨在为体育观众挑选精彩场景。我们在名为 FightingICE 的格斗游戏平台上开展了用户研究，结果表明，所提方法生成的游戏过程比标准蒙特卡洛树搜索生成的游戏过程更具娱乐性。FightingICE [6],自2013年以来一直用于国际游戏人工智能竞赛，包括 CoG 2019 上的一项竞赛。所提出的引入精彩片段线索至蒙特卡洛树搜索的评估函数中以生成娱乐性游戏过程的思路，也可应用于其他游戏，前提是具备游戏前向模型，并对线索进行相应调整。

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II. 相关工作

A. 格斗游戏中的蒙特卡洛树搜索

尽管近年来已出现使用深度学习技术的人工智能 [7-9]，但蒙特卡洛树搜索仍是实现格斗游戏人工智能的一种常用技术，它结合了蒙特卡洛方法和利用给定前向模型的博弈树搜索。一个在 FightingICE 中使用蒙特卡洛树搜索的示例人工智能 [10] 采用了开环方法 [11]。在本研究中，我们沿用这一方案。

图 1 展示了蒙特卡洛树搜索的概览。在开环方法中，除根节点外，每个节点都表示一个动作；根节点则表示当前游戏状态，该状态由各角色的生命值（HP）、能量、坐标和动作以及游戏剩余时间等信息定义。一条边表示所关注动作的持续执行。该方法包含四个步骤：选择、扩展、模拟和反向传播。下文将分别介绍这些步骤。

1) 选择：根据所采用的选择准则，从根节点开始选择节点，直至到达叶节点。我们使用广泛应用于此任务的置信上界（$UCB1$） [12], ，其定义如下式所示：

$$UCB1_i = \overline{X}_i + C\sqrt{\frac{2\ln N}{N_i}} \quad (1)$$

其中， N_i 为节点（动作） i 被访问的次数， N 为其父节点的访问次数， C 为常数。此外， X_i 为节点 i的平均评估值：

$$\overline{X}_i = \frac{1}{N_i}\sum_{j=1}^{N_i} Eval_j \quad (2)$$

其中， $Eval_j$ 是一个评估函数，用于从人工智能玩家的视角返回第 j次模拟中获得的奖励值。请注意，树中的每个节点都包含 $UCB1$值和一个用于统计该节点被访问次数的计数器。在本研究中，选定路径包含

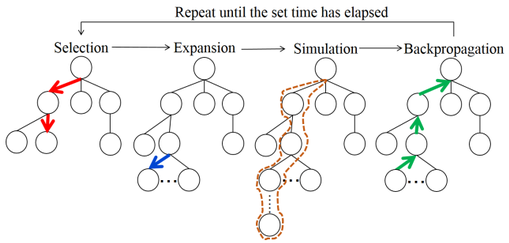


Fig. 1. An overview of MCTS for a fighting game

the nodes with the highest $UCB1$ value, from the root node until a leaf node.

2) Expansion: After a leaf node has been reached in the $Selection$ step, if the number of times it has been explored exceeds a threshold N_{max} and the depth of the tree is lower than a threshold D_{max}, all of its child nodes are created at once from it. Note that the initial tree consists of the root node and all of its direct child nodes.

3) Simulation: A simulation of the length of T_{sim} is performed by sequentially executing all the actions in the selected path while the opponent's actions are chosen for execution at random. If T_{sim} has not passed yet after those actions of the AI player have been executed, a rollout will be carried out until T_{sim} runs out using randomly selected actions. At the end of each simulation, the character's $Eval_j$ is calculated.

4) Backpropagation: The value of $Eval_j$ obtained in the $Simulation$ step is back-propagated from the leaf node to the root node during which the $UCB1$ value at each node along the path is also updated accordingly.

MCTS repeats the above four steps until a given time limit, T_{max}, is reached. An action is then chosen among all of the direct child nodes of the root node as the next action according to a given recommendation policy. In this work, it is action i^* that has the highest $\overline{X}_i$. As done in the open-loop approach, the chosen child node will be used as the next root node, now representing the state after the action is executed, and its sibling nodes will be pruned.

B. Highlight Generation

With a rapid increase in sports broadcasting, it is necessary to generate a highlight that allows audiences to see exciting scenes at their convenient timing. However, manual generation of highlights is time consuming, so a number of automatic methods for generating highlights have been proposed. Typically, scenes are evaluated based on several cues, and scenes with high evaluation values are selected for a highlight. For example, for boxing, a highlight can be generated based on the camera-flash timing and the distance between both players [13], i.e., a set of scenes with a close distance during flash light becomes a highlight.

Recently, a method was proposed that generates a highlight for basketball based on five cues [14]: "Audio," "Score Differential," "Player Ranking," "Basket Type," and "Motion," described in the following. Audio assesses scenes according to the loudness of spectators and commentators. Score Differential considers that scenes with a narrow gap in score near the end of the game are exciting. Player Ranking selects scenes where a shot is done by a high-ranking player. Basket Type ranks scenes according to their scoring shot types. We select and modify certain cues from these two previous studies for our work; worth mentioning is more recent work by Ringer and Nicolaou [15] that takes into account information on streamers' face and audio, which cannot be applied to our work.

III. Proposed Method

In this section, we describe our proposed method for generating gameplay to entertain spectators. Our research hypothesis is that a fight between two AI players would be exciting if each AI player selects their actions based on highlight cues, which in theory would make their fight a highlight from the beginning to the end. In this work, MCTS is used, and three highlight cues are introduced to form the evaluation function of MCTS: "Score Transition," "Action," and "Distance." All of them are expected to increase aggression in gameplay, which is what spectators demand in both traditional sports [16] and esports [17]. The details of the proposed method are given below.

A. Score Transition

Score Transition is a term that combines the elapsed time of the current round and the predicted damage value that the AI player gives to its opponent. This term, $Score$, is defined as follows:

$$Score = RoundTime \times Damage \tag{3}$$

where $RoundTime$ and $Damage$ represent the elapsed fight time and the damage value at the end of the current simulation, respectively. According to this term, the AI player prioritizes actions with high damage values, and this kind of an earnest fight is more prominent as the time is closer to the end of the round. The term is based on Score Differential [14], but has been adapted to the fighting game accordingly.

B. Action

Action prioritizes certain actions by the AI player and is defined as follows:

$$Action = \begin{cases} \frac{1}{2^{Rank-1}} & (belongs\ to\ RankAct) \\ 0 & (otherwise) \end{cases} \tag{4}$$

where $RankAct$ is a list of actions, and $Rank$ is the value associated to each action in the list. The list of actions and their rank are shown in Table I. In previous work [14], basket shots (dunk, three-point jumper, etc.) that have a high degree of excitement were selected to form the Basket Type cue. In this work, the actions in the list are those we consider have high visual effects. The value of an action that is not in the list is 0.

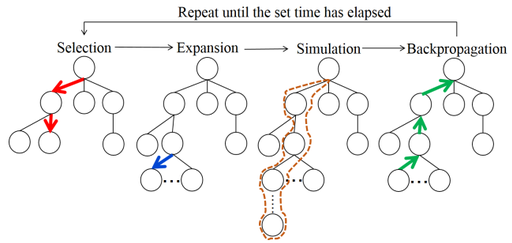


图 1. 格斗游戏中的蒙特卡洛树搜索概览

从根节点到叶节点之间具有最高 $UCB1$ 值的节点。

2) 扩展：在$Selection$ 步骤中到达叶节点后，如果该节点被探索的次数超过阈值 N_{max} ，且树的深度低于阈值 D_{max}，则一次性创建该节点的所有子节点。请注意，初始树由根节点及其所有直接子节点组成。

3) 模拟：通过依次执行选定路径中的所有动作，进行长度为 T_{sim} 的模拟，同时随机选择并执行对手的动作。如果在执行完人工智能玩家的这些动作后， T_{sim} 尚未结束，则使用随机选择的动作进行模拟推演，直到 T_{sim} 耗尽。每次模拟结束时，计算角色的 $Eval_j$。

4) 反向传播：将在$Simulation$ 步骤中获得的 $Eval_j$ 值从叶节点反向传播至根节点，同时相应地更新该路径上每个节点的 $UCB1$ 值。

蒙特卡洛树搜索重复上述四个步骤，直到达到给定的时间限制T_{max}。然后，根据给定的推荐策略，从根节点的所有直接子节点中选择一个动作作为下一个动作。在本研究中，所选动作是具有最高 X_i的动作 i^*。与开环方法一样，所选子节点将用作下一个根节点，此时它表示该动作执行后的状态，而其兄弟节点将被剪除。

B. 精彩片段生成

随着体育转播的迅速增多，有必要生成精彩集锦，让观众能够在方便的时候观看精彩场面。然而，人工制作精彩集锦耗时较长，因此，研究者提出了多种自动生成精彩集锦的方法。通常，这些方法根据若干线索对场景进行评估，并选择评估值较高的场景组成精彩集锦。例如，对于拳击比赛，可以根据相机闪光时机和双方选手之间的距离生成精彩集锦 [13], ，即将闪光时双方距离较近的一组场景组成精彩集锦。

最近，有研究提出了一种基于五种线索生成篮球精彩集锦的方法 [14]: "音频" "得分差" "球员排名" "投篮类型" 和 "运动"，具体说明如下。音频根据观众和解说员的响度来评估场景。得分差认为，比赛临近结束时得分差距较小的场景令人兴奋。球员排名选取排名靠前的球员投篮的场景。投篮类型根据得分投篮类型对场景进行排序。我们从上述两项研究中选取了某些线索并加以修改，用于本研究；值得一提的是，Ringer 和 Nicolaou 在较近期的研究 [15]中考虑了主播的面部和音频信息，但这些信息无法应用于本研究。

III. 所提方法

本节介绍我们提出的用于生成具有观赏性的游戏过程的方法。我们的研究假设是，如果两个人工智能玩家各自依据精彩片段线索选择动作，那么它们之间的战斗就会十分精彩，理论上，这将使整场战斗从头到尾都成为精彩集锦。本研究采用蒙特卡洛树搜索，并引入三种精彩片段线索来构建蒙特卡洛树搜索的评估函数："得分变化" "动作" 和 "距离" 。这些线索均旨在增强游戏过程中的进攻性，而这正是传统体育 [16] 和电子竞技 [17]观众共同的需求。下文将详细介绍所提方法。

A. 得分变化

得分变化是一个结合当前回合的已用时间与人工智能玩家对对手造成的预测伤害值的项。该项 $Score$定义如下：

$$Score = RoundTime \times Damage \tag{3}$$

其中， $RoundTime$ 和 $Damage$ 分别表示当前模拟结束时的已用战斗时间和伤害值。根据该项，人工智能玩家会优先选择伤害值较高的动作，而且越接近回合结束，这种全力以赴的战斗表现就越突出。该项以得分差 [14], 为基础，并针对格斗游戏进行了相应调整。

B. 动作

动作优先考虑人工智能玩家的某些动作，其定义如下：

$$Action = \begin{cases} \frac{1}{2^{Rank-1}} & (belongs\ to\ RankAct) \\ 0 & (otherwise) \end{cases} \tag{4}$$

其中， $RankAct$ 是动作列表， $Rank$ 是与列表中每个动作关联的数值。动作列表及其排名如表 I 所示。在先前的研究中， [14], 选取了令人高度兴奋的投篮（扣篮、三分跳投等）来构成投篮类型线索。在本研究中，列表中的动作是我们认为具有强烈视觉效果的动作。不在列表中的动作，其数值为 0。

TABLE I
RankAct LIST IN FIGHTINGICE

RankAct	Skill content	*Rank*
STAND_D_DF_FC	Special Skill	1
STAND_F_D_DFB	Strong Upper	2
STAND_D_DB_BB	Sliding Kick	3
STAND_D_DF_FB	Shoot Strong Projectile Forward	4

C. Distance

Distance prioritizes a close-distance fight near the center of the screen and is defined as follows:

$$Distance = 1 - \left| \frac{center - Xpos}{center} \right| \quad (5)$$

where $center$ is the x coordinate at the center of the screen, and $Xpos$ is the AI player's x coordinate. $Distance$ will have a higher value when the AI player is positioned closer to the center of the screen. Note that if the distance between both AI players is only used, they might end up fighting at either edge of the screen, which we consider less exciting.

D. Evaluation Function

Finally, the evaluation function of MCTS in Eqn. (2) is concretely defined as follows:

$$Eval_j = \omega_s Score + \omega_a Action + \omega_d Distance \quad (6)$$

where $Score$, $Action$, and $Distance$ are normalized here to the range of 0 and 1, and their weights $\omega_s, \omega_a, \omega_d$ are discussed in the next section.

IV. EXPERIMENTS

In this section, we describe experiments conducted to verify the performance of the proposed method. We compare a number of AIs, which use a different combination of the three cues described above, with a typical MCTS AI (mAI) [10], which is a sample MCTS AI for the aforementioned competition using FightingICE where the difference in the HP decrease between the player AI and its opponent is used in the evaluation function. Results from a user study and gameplay analysis are given.

A. Environment

1) FightingICE: FightingICE is a real-time 2D fighting game platform used in a game AI competition (FTGAIC)[1] and for research [18-24, 26]. FightingICE has been originally developed from scratch without using a ROM emulator and publicly made available. As a result, use of this platform imposes no infringement issues to be concerned, in particular for places where there are no explicit law doctrines similar to $fair\ use$ in the US.

In FightingICE, a round lasts 60 seconds, and the game is rendered 60 frames per second. Due to receiving a delayed game state from the system, an AI player does not precisely know the timing that it can perform its next action and hence it has to decide and input an action every frame, by which the previous action awaiting execution will be overridden. The HP for both characters is initially set to HP_{max} and decreases when a character of interest is hit. A round ends when the fight is conducted for 60 seconds or the HP of at least one of the two characters becomes 0. The player of the character with the larger remaining HP at the end of a round is the round's winner. In our experiments, the value of HP_{max} was set to 400 according to the rule of the Standard Track of FTGAIC.

In FTGAIC, the aforementioned delay of the current game state provided to AI players is a constraint that has been imposed. More specifically, both AI players can only obtain each time a game state delayed by 15 frames (0.25 s), taking into account the delay of human perception. However, the delay constraint was removed in the experiments. This is because our work is focused on generation of entertaining gameplay, not on development of an AI player for fighting against another AI opponent in a competition or a human opponent in a fair fashion.

2) Parameter Settings: In the experiments, we used five AI players: mAI, AI1, AI2, AI3, and AI4. The last four AI players used a different combination of cues in the evaluation function as can be seen by the values of ω_s, ω_a, and ω_d shown in Table II. The values of other parameters in use are summarized in Table III where the first five parameters, determined following our previous work [5] except for C being set to 1 which is a typical value when the evaluation function is normalized, were shared among all the AI players and the sixth, $center$, was only for those AI players that included $Distance$ in the evaluation function, i.e., A1, A2, and A3.

TABLE II
WEIGHTS IN THE EVALUATION FUNCTION

Name	ω_s	ω_a	ω_d
AI1	1/3	1/3	1/3
AI2	0	1/2	1/2
AI3	1/2	0	1/2
AI4	1/2	1/2	0

B. User Study

In this user study, we evaluated the fun of gameplay generated by the aforementioned five AI players[2]. Participants were mainly college students and/or followers of the Facebook

TABLE III
PARAMETERS USED IN THE EXPERIMENTS

Notation	Description	Value
C	Balancing parameter	1
N_{max}	Threshold of the number of visits	10
D_{max}	Threshold of the tree depth	10
T_{sim}	Simulation-time budget	60 frames
T_{max}	Execution time of MCTS	16.5 ms
$center$	Center of the game screen	480 pixels

[1]http://www.ice.ci.ritsumei.ac.jp/%7eftgaic/
https://github.com/TeamFightingICE/FightingICE

[2]http://www.ice.ci.ritsumei.ac.jp/%7eruck/hlmcts-cog2019.htm

表 I*RankAct*FightingICE 中的列表

RankAct	技能内容	*Rank*
STAND_D_DF_FC	特殊技能	1
STAND_F_D_DFB	强力上勾拳	2
STAND_D_DB_BB	滑踢	3
STAND_D_DF_FB	向前发射强力飞行道具	4

C. 距离

距离优先考虑屏幕中心附近的近距离战斗，其定义如下：

$$Distance = 1 - \left| \frac{center - Xpos}{center} \right| \quad (5)$$

其中， $center$ 为屏幕中心的 x 坐标， $Xpos$为人工智能玩家的 x 坐标。人工智能玩家的位置越靠近屏幕中心， $Distance$ 的值就越高。请注意，如果仅使用两个人工智能玩家之间的距离，它们最终可能会在屏幕任一侧的边缘交战，而我们认为这样的对战不够精彩。

D. 评估函数

最后，式（2）中的蒙特卡洛树搜索评估函数具体定义如下：

$$Eval_j = \omega_s Score + \omega_a Action + \omega_d Distance \quad (6)$$

其中， $Score$、 $Action$和 $Distance$ 在此被归一化至0到1的范围，其权重 ω_s、 ω_a、 ω_d 将在下一节讨论。

IV. 实验

本节介绍为验证所提方法的性能而开展的实验。我们将采用上述三种线索的不同组合的多个人工智能，与一个典型的蒙特卡洛树搜索人工智能（mAI）[10],进行比较。该人工智能是前述使用FightingICE的竞赛所提供的蒙特卡洛树搜索人工智能示例，其评估函数使用玩家人工智能与对手之间的生命值减少量之差。本节给出了用户研究和游戏过程分析的结果。

A. 环境

1) FightingICE：FightingICE 是一个实时二维格斗游戏平台，用于游戏人工智能竞赛（FTGAIC）[1]和研究 [18-24, 26]。FightingICE 最初从零开始开发，未使用ROM模拟器，并已向公众开放。因此，使用该平台无需担心侵权问题，尤其是在没有类似美国$fair\ use$ 的明确法律原则的地区。

在 FightingICE 中，每个回合持续 60 秒，游戏以每秒帧数为 60 的速率渲染。由于从系统接收到的游戏状态存在延迟，人工智能玩家无法准确地知道何时可以执行下一个动作，因此必须在每一帧决定并输入一个动作，这会覆盖之前等待执行的动作。两个角色的生命值最初均设为 HP_{max}，并在相应角色受到攻击时减少。当对战持续 60 秒，或两个角色中至少一个角色的生命值降至 0 时，回合结束。回合结束时，剩余生命值较高的角色所对应的玩家获胜。在我们的实验中，根据 FTGAIC 标准赛道的规则， HP_{max} 的值设为 400。

在 FTGAIC 中，上述向人工智能玩家提供的当前游戏状态的延迟是一项强制约束。更具体地说，考虑到人类感知的延迟，双方人工智能玩家每次都只能获取延迟了 15 帧（0.25 秒）的游戏状态。然而，实验中取消了这一延迟约束。这是因为我们的工作重点是生成娱乐性游戏过程，而不是开发用于在竞赛中对抗另一个人工智能对手，或与人类对手公平对战的人工智能玩家。

2) 参数设置：实验中，我们使用了五个人工智能玩家：mAI、AI1、AI2、AI3 和 AI4。后四个人工智能玩家在评估函数中使用了不同的线索组合，具体可见表 II 中所示的 ω_s、 ω_a 和 ω_d 的取值。所用其他参数的取值汇总于表 III。其中，前五个参数由所有人工智能玩家共用，其取值参照我们之前的研究 [5] 确定，但 C 设为 1，这是评估函数归一化时的典型值；第六个参数 $center$ 仅用于在评估函数中包含 $Distance$ 的人工智能玩家，即 A1、A2 和 A3。

表 II 评估函数中的权重

Name	ω_s	ω_a	ω_d
AI1	1/3	1/3	1/3
AI2	0	1/2	1/2
AI3	1/2	0	1/2
AI4	1/2	1/2	0

B. 用户研究

在本次用户研究中，我们评估了上述五个人工智能玩家[2]所生成的游戏过程的趣味性。参与者主要是大学生和/或 Facebook 上的关注者

[2]http://www.ice.ci.ritsumei.ac.jp/%7eruck/hlmcts-cog2019.htm

表 III 实验中使用的参数

符号	说明	值
C	平衡参数	1
N_{max}	访问次数阈值	10
D_{max}	树深度阈值	10
T_{sim}	模拟时间预算	60 帧
T_{max}	蒙特卡洛树搜索的执行时间	16.5 毫秒
$center$	游戏画面中心	480 像素

site of FTGAIC. They were asked to watch two gameplay video clips: a gameplay video clip where both AI players are of the mAI type and the other gameplay video clip where both AI players are of the same type selected from one of the other four AI players. Then, given no information on how each gameplay was generated, they were asked to answer which gameplay was more fun and to comment their reasons for selection. Three rounds were generated by two AI players of the same type, and the content in the last 30 seconds of a round was recorded. The user study was conducted on an online survey site which displayed a pair of gameplay video clips of the same but randomly selected round number. To test statistical differences between any two methods, the exact binomial test was used on amalgamated results from all three rounds.

Results and Discussions: Fig. 2 shows the results of the user study. In the first half of our study (Fig. 2-(1)) where we compared gameplay by mAI and gameplay by AI1, 66 participants (57 males, 9 females), with an average age of 24.1 ± 2.5, participated. In general, the majority of them preferred the AI1 gameplay for all rounds, indicating the positive effect of using all of Score Transition, Action, and Distance. The effectiveness of these terms can also be seen in representative comments by the participants in Table IV. A difference at the 1% level of significance ($p = 0.002$) was found between the gameplay of these two AI-player types. As a result, it can be said that the AI1 gameplay is more fun than the mAI gameplay.

TABLE IV
REASONS FOR CHOICE

AI	Reason
$AI1$	Frequent changes of the dominance side More engaging gameplay Closer-range fight
mAI	Well-thought out gameplay Curious to know how the fight will be unfolded Have both close-range and far-range fights

In the second half of our study conducted after the first one, we examined how the user evaluation toward gameplay would change if one of Score Transition, Action, or Distance was not included in the evaluation function, i.e., AI2 gameplay, AI3 gameplay, and AI4 gameplay, respectively. Our results indicated that the mAI gameplay was not statistically different from the AI2 gameplay and the AI4 gameplay, but statistically different from the AI3 gameplay at the 5% level of significance. For statistical significance tests, the number of participants was adjusted in each test to 66, the number of participants in the first half of the user study, for fair comparison. Below are details on the comparisons between mAI and AI2, mAI and AI3, and mAI and AI4. Note that due to random assignment of participants to one of the three comparisons, their numbers of participants were different and that in statistical tests, a significant difference can be more readily obtained with a higher number of participants.

The results of the comparison between the mAI gameplay and the AI2 gameplay are shown in Fig. 2-(2) where 17 participants (14 males, 3 females), with an average age of 23.1 ± 1.8, answered. For Round 2, all of the participants watching this round answered that the AI2 gameplay was more fun, but vice versa for Round 1. In Round 3, the number of participants who preferred either gameplay was the same. Due to the absence of Score Transition in AI2, a close-range fight using the actions in the $RankAct$ list (Table I) was continuously performed in the center of the screen throughout the fight. Therefore, the fight was sometimes repetitive causing the participants to not enjoy the gameplay. The exact binomial test, even after adjusting the number of participants to 66, showed no significant difference between the mAI gameplay and the AI2 gameplay. As a result, it can be said that the fun of the gameplay by these two AI player types is not different; in other words, Score Transition is an important element in the evaluation function.

Fig. 2-(3) shows the results of the comparison between the mAI gameplay and the AI3 gameplay where another group of 17 participants (14 males, 3 females), with an average age of 23.6 ± 2.1, participated. In Rounds 1 and 2, according to answers by participants the AI3 gameplay was more fun, but in Round 3, both types of gameplay had the same number of participants who preferred either of them. These results were obtained because there was no Action in AI3. In the gameplay by both AI3 players, a fight occurred near the center of the screen and became more tensed as the fight time increased, but the frequency of using actions, with high visual effects, in $RankAct$ decreased. The exact binomial test showed that there was a significant difference between the mAI gameplay and the AI3 gameplay at the 5% level of significance ($p = 0.019$) after the number of participants was adjusted to 66. Accordingly, it can be said that the AI3 gameplay is more fun than the mAI gameplay, but with less reliable than the difference between the mAI gameplay and the AI1 gameplay.

In the comparison between the mAI gameplay and the AI4 gameplay, 10 participants (9 males, 1 females), with an average age of 23.3 ± 1.8, participated, and the results are shown in Fig. 2-(4). In Round 2, many participants answered that the AI4 gameplay was more fun, but in Round 1, the trend became opposite. In Round 3, both types of gameplay had the same number of votes. This is because there was no Distance in AI4. Due to the absence of it, each of the AI4 players used actions in $RankAct$ throughout the round, causing a lot of damage to the opponent especially in the second half of the round, but they could not reduce the distance to the opponent. As a result, long-distance attacks were repeated, so there were not many changes in the fight causing the participants to find it not so fun. The exact binomial test, after the number of participants was adjusted to 66, showed no significant difference between both types of gameplay. From the above results, it can be said that the fun of both gameplay is not different; in other words, Distance is an important element in the evaluation function.

From all of the above results, it can be said that the AI1 gameplay where all the cue elements are used is more fun than

，他们关注的是 FTGAIC 的页面。参与者被要求观看两个游戏过程视频片段：其中一个片段中的两个人工智能玩家均为 mAI 类型，另一个片段中的两个人工智能玩家则均为从其余四个人工智能玩家中选出的一种相同类型。在未获知各段游戏过程如何生成的情况下，参与者随后被要求回答哪段游戏过程更有趣，并说明其选择理由。由两个相同类型的人工智能玩家进行三个回合的游戏，并录制每个回合最后 30 秒的内容。本次用户研究通过在线调查网站开展，该网站展示一对游戏过程视频片段，两段视频的回合编号相同，但该编号为随机选取。为检验任意两种方法之间的统计差异，我们对三个回合的汇总结果进行了精确二项检验。

结果与讨论：图 2 展示了用户研究的结果。在研究的前半部分（图 2-(1)），我们比较了 mAI 和 AI1 的游戏过程，共有 66 名参与者（57 名男性、9 名女性）参加，平均年龄为24.1 ± 2.5。总体而言，在所有回合中，大多数参与者都更喜欢 AI1 的游戏过程，这表明同时使用得分变化、动作和距离这三项指标具有积极作用。表 IV 中参与者的代表性评论也体现了这些指标的有效性。这两种 AI 玩家类型的游戏过程之间存在显著差异，达到 1% 的显著性水平（$p = 0.002$）。因此，可以说 AI1 的游戏过程比 mAI 的游戏过程更有趣。

表 IV 选择理由

AI	理由
$AI1$	占优势的一方频繁变化 游戏过程更引人入胜 交战距离更近
mAI	经过深思熟虑的游戏过程 好奇战斗将如何展开 兼有近距离和远距离战斗

在第一阶段结束后开展的研究后半部分中，我们考察了当评估函数不包含得分变化、动作或距离中的某一项时，用户对游戏过程的评价会如何变化，这三种情况分别对应 AI2 、 AI3 和 AI4 的游戏过程。结果表明，在 5% 的显著性水平下，mAI 的游戏过程与 AI2 和 AI4 的游戏过程均无统计显著性差异，但与 AI3 的游戏过程存在统计显著性差异。为进行公平比较，在各项统计显著性检验中，参与者人数均调整为 66 人，即用户研究前半部分的参与者人数。下文详细介绍 mAI 与 AI2、mAI 与 AI3 以及 mAI 与 AI4 之间的比较。请注意，由于参与者被随机分配至三组比较中的一组，各组的参与者人数不同；而在统计检验中，参与者人数越多，就越容易得出显著差异。

mAI 游戏过程的比较结果

与 AI2 游戏过程的比较结果如图 2-(2) 所示，共有 17 名参与者（14 名男性、3 名女性）作答，平均年龄为 23.1 ± 1.8。对于第 2 回合，所有观看该回合的参与者均表示 AI2 的游戏过程更有趣，而第 1 回合的结果则恰好相反。在第 3 回合中，偏好两种游戏过程的参与者人数相同。由于 AI2 中没有得分变化，整个战斗过程中，双方始终在屏幕中心使用 $RankAct$ 列表（表 I）中的动作进行近距离战斗。因此，战斗有时显得重复，导致参与者无法从游戏过程中获得乐趣。即使将参与者人数调整为 66 人，精确二项检验仍显示，mAI 的游戏过程与 AI2 的游戏过程之间没有显著差异。因此，可以认为这两种 AI玩家类型所呈现的游戏过程在趣味性上没有差异；换言之，得分变化是评估函数中的重要元素。

图2-(3)展示了mAI游戏过程与AI3游戏过程的比较结果，另有一组17名参与者（14名男性、3名女性）参与了此次比较，平均年龄为 23.6 ± 2.1。根据参与者的回答，在第1和第2回合中，AI3游戏过程更有趣，但在第3回合中，偏好这两种游戏过程的参与者人数相同。之所以得到这些结果，是因为AI3中没有动作。在双方均由AI3操控的游戏过程中，战斗发生在屏幕中心附近，并随着战斗时间的增加而变得更加紧张，但在$RankAct$ 中，使用具有强烈视觉效果的动作的频率有所下降。将参与者人数调整为66后，精确二项检验表明，mAI游戏过程与AI3游戏过程在5%的显著性水平下存在显著差异（$p = 0.019$）。因此，可以说AI3游戏过程比mAI游戏过程更有趣，但这一结论的可靠性低于mAI游戏过程与AI1游戏过程之间差异的可靠性。

在 mAI 游戏过程与 AI4 游戏过程的比较中，共有 10 名参与者（9 名男性、1 名女性），平均年龄为 23.3 ± 1.8，结果如图 2-(4) 所示。在第 2 回合中，许多参与者认为 AI4 的游戏过程更有趣，但在第 1 回合中，趋势则相反。在第 3 回合中，两种游戏过程获得的票数相同。这是因为 AI4 中没有距离这一要素。由于缺少这一要素，各个 AI4 玩家在整个回合中都使用 $RankAct$ 中的动作，给对手造成了大量伤害，尤其是在回合的后半段，但它们无法缩短与对手的距离。因此，远距离攻击不断重复，战斗变化较少，使参与者觉得不太有趣。将参与者人数调整为 66 后进行的精确二项检验表明，两种游戏过程之间没有显著差异。根据上述结果，可以认为两种游戏过程的趣味性没有差异；换言之，距离是评估函数中的一个重要要素。

综合上述所有结果可以说，使用全部提示元素的 AI1 游戏过程比

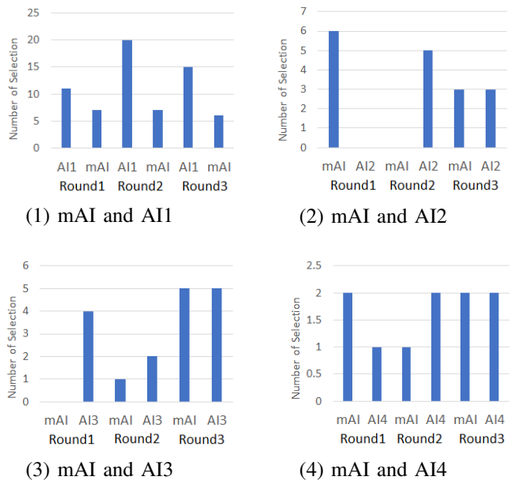


Fig. 2. Results of the user study

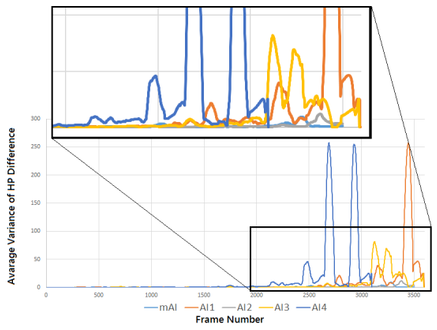


Fig. 3. The average variance of HP difference where the x-axis shows the frame number and the y-axis shows the average variance

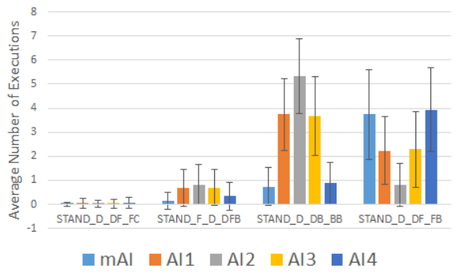


Fig. 4. The average use of $RankAct$ actions

the mAI gameplay. Although when the evaluation function does not include Action, the AI3 gameplay outperforms the mAI gameplay with a statistical significance, inclusion of Action with the other cue elements contributes to a statistical significance with a more reliability. As a result, all of the proposed three cue elements forming the evaluation function are important to the fun of generated gameplay.

C. Gameplay Analyses

Here, we performed an analysis of gameplay generated by mAI, AI1, AI2, AI3, and AI4. In particular, we generated 1000 rounds of gameplay for each type and analyzed them using three criteria: "the Average Variance of the HP Difference" (AVHPD), "the Average Use of $RankAct$ Actions" (AURAA), and "the Average Distance Between the Two Characters" (ADBTC) defined as follows:

- AVHPD: the average among 1000 rounds of the variance of the HP difference between the two AI players among each 100 consecutive frames
- AURAA: the average among 1000 rounds of the number of times that each action in $RankAct$ is executed per one AI player throughout its fight
- ADBTC: the average among 1000 rounds of the average distance between the two characters throughout their fight

Results and Discussions: AVHPD for gameplay by each AI-player type is shown in Fig. 3. It can be seen that in the gameplay generated by the three AI-player types that use Score Transition (AI1, AI3, AI4), the HP difference between the two AI players varies more swiftly, causing a higher variance, near the end of round. The finding substantiates our discussions on Score Transition in the previous sub-section.

AURAA for gameplay by each AI-player type is shown in Fig. 4 in the form of histograms with error bars. It can be seen that, due to an emergent behavior governed by each term in the evaluation function, AI1 performed the 2nd and 3rd ranked actions in the $RankAct$ list more frequently than mAI although the latter performed the 4th ranked action more frequently than the others except for AI4. This result in part substantiates the superiority of AI1 over mAI in the user study. However, the 1st ranked action in $RankAct$, the one with the highest visual effect which requires a large amount of energy to perform, was rarely executed by AI1 and its variants. As a result, there is room for improvement, e.g., how to promote execution of this kind of action, which is left as our future work.

ADBTC is shown in Fig. 5. It can be clearly seen that the gameplay generated by AI1, AI2, and AI3 that include Distance in the evaluation function represents a close-distance fight. In addition, mAI and AI4 with no Distance in the evaluation function generate fights where the two characters keep a further distance between each other. This finding shows that Distance successfully constrains the distance between the characters during their fight.

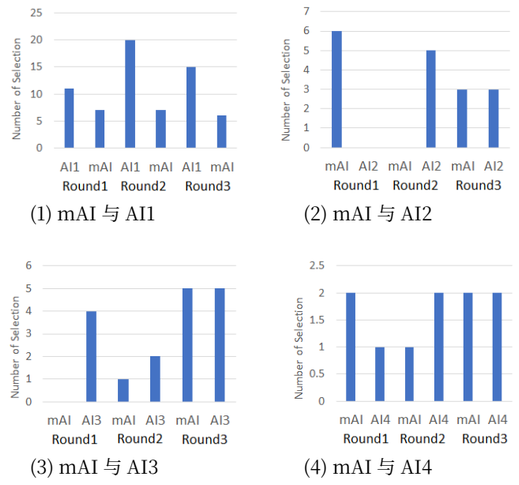


图 2. 用户研究结果

mAI 游戏过程更有趣。尽管在评估函数不包含动作时，AI3 游戏过程优于 mAI 游戏过程，且差异具有统计显著性，但将动作与其他线索要素一同纳入评估函数，有助于获得更可靠的统计显著性。因此，所提出的构成评估函数的三个线索要素，对于所生成游戏过程的趣味性都很重要。

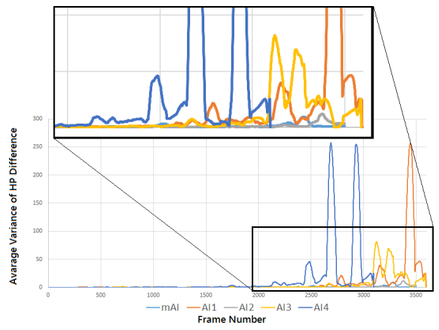


图 3. 生命值差的平均方差，其中横轴表示帧编号，纵轴表示平均方差

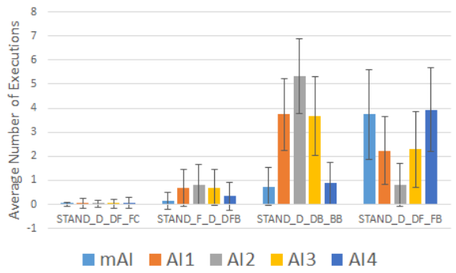


图4. $RankAct$ 动作的平均使用情况

C. 游戏玩法分析

在此，我们对 mAI、AI1、AI2、AI3 和 AI4 生成的游戏过程进行了分析。具体而言，我们为每种类型生成了 1000 个回合的游戏过程，并采用以下三个指标进行分析：“生命值差的平均方差”（AVHPD）、“ $RankAct$ 动作的平均使用次数”（AURAA）和“两个角色之间的平均距离”（ADBTC），其定义如下：

- AVHPD：在每连续 100 帧中计算两个人工智能玩家之间生命值差的方差，再对 1000 个回合取平均值
- AURAA：单个人工智能玩家在整个对战过程中执行 $RankAct$ 中每个动作的次数在 1000 个回合中的平均值
- ADBTC：两个角色在整个对战过程中的平均距离在 1000 个回合中的平均值

结果与讨论：图 3 展示了各 AI 玩家类型所生成游戏过程的生命值差的平均方差。可以看出，在使用得分变化的三种 AI 玩家类型（AI1、AI3、AI4）生成的游戏过程中，两个 人工智能玩家之间的生命值差在回合接近结束时变化更为迅速，从而产生更高的方差。这一发现证实了我们在上一小节中关于得分变化的讨论。

图 4 以带误差棒的直方图形式展示了各 AI 玩家类型所生成游戏过程的 AURAA。可以看出，由于评估函数中各项所支配的涌现行为，AI1 执行 $RankAct$ 列表中排名第二和第三的动作的频率高于 mAI，尽管后者执行排名第四的动作的频率高于除 AI4 以外的其他类型。这一结果在一定程度上证实了用户研究中 AI1 优于 mAI 的结论。然而，AI1 及其变体很少执行 $RankAct$ 中排名第一的动作，即视觉效果最强但需要消耗大量能量才能执行的动作。因此，仍有改进空间，例如如何促进此类动作的执行，这将留待我们未来研究。

ADBTC 如图 5 所示。可以清楚地看出，在评估函数中纳入距离的 AI1、AI2 和 AI3 所生成的游戏过程呈现为近距离战斗。此外，评估函数中未纳入距离的 mAI 和 AI4 所生成的战斗中，两个角色彼此保持较远的距离。这一发现表明，距离成功地约束了角色在战斗中的间距。

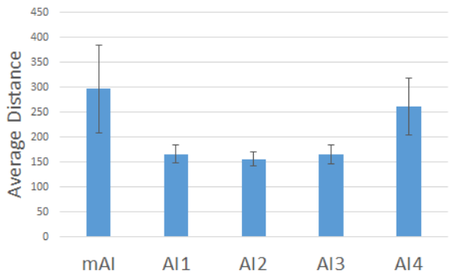


Fig. 5. The average distance between the two characters

V. Conclusions and Future Work

In this paper, we proposed a fighting game AI with the aim to generate entertaining gameplay. Based on existing highlight generation methods, we introduced three cues for use in the evaluation function of MCTS. Our results from the conducted user study showed that the proposed method that has all of the three highlight cues, AI1, generated gameplay that was more fun than gameplay generated by an existing MCTS AI that aims at decreasing its opponent's HP while maintaining its own HP. These results were also substantiated by the conducted gameplay analyses.

However, one issue is left as future work that is how to have an AI player of interest execute an action with a high visual effect that consumes a large amount of energy. Active learning for parameter tuning [25] might be useful. We are also interested in generation of believable gameplay [26-27].

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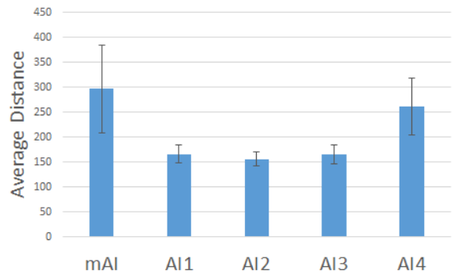


图5. 两个角色之间的平均距离

V. 结论与未来工作

本文提出了一种旨在生成娱乐性游戏过程的格斗游戏人工智能。基于现有的精彩片段生成方法，我们引入了三种线索，用于蒙特卡洛树搜索的评估函数。用户研究的结果表明，包含全部三种精彩片段线索的所提方法 AI1 所生成的游戏过程，比现有蒙特卡洛树搜索人工智能生成的游戏过程更有趣；后者旨在降低对手生命值的同时维持自身生命值。所开展的游戏玩法分析也证实了这些结果。

然而，仍有一个问题有待未来研究解决，即如何让所关注的人工智能玩家执行消耗大量能量且具有强烈视觉效果的动作。用于参数调优的主动学习 [25]或许会有所帮助。我们还对生成可信的游戏过程 [26-27]感兴趣。

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